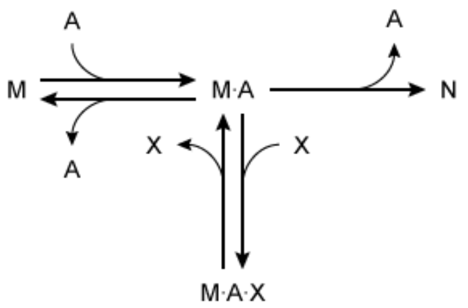
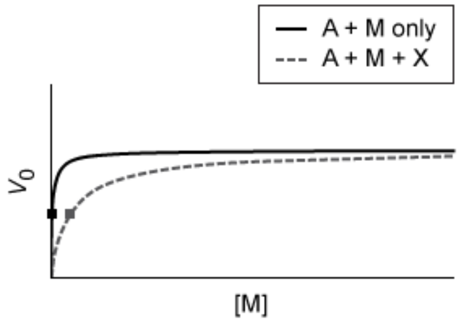
A monomeric enzyme A catalyzes the reaction of substrate M to product N. A novel compound X reversibly interacts with this enzyme-catalyzed reaction according to the reaction scheme below.



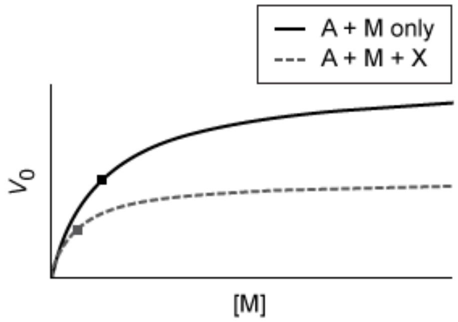
Which of the following best represents a Michaelis-Menten graph demonstrating the most likely effects of compound X on the kinetics of enzyme A? Note: the point on each graph where V_0 = ½ V_{max} is marked with a square.

A.



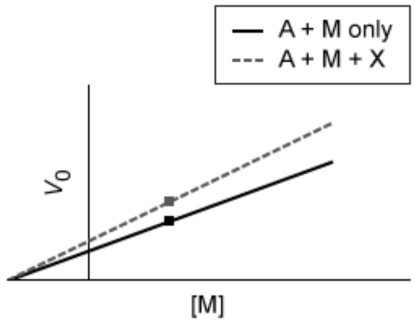
[18%]

B.



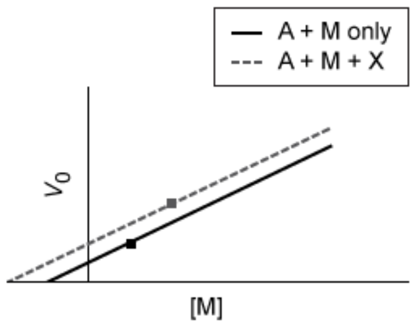
[56%]

C.



[10%]

D.



[14%]

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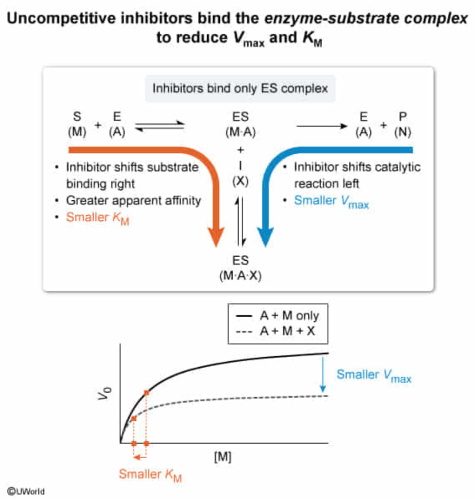
Correct
55% answered correctly

Explanation:



The reversible inhibition of enzymes can be classified by which species the inhibitors bind. The reaction scheme shows that inhibitor X binds only to the complex formed between the enzyme A and the substrate M (M·A). Binding *only* to the **enzyme-substrate complex** is a property of uncompetitive inhibitors. By binding to this complex, uncompetitive inhibitors can prevent the catalytic formation and subsequent release of product, **lowering the apparent maximum velocity V_{max}**. Additionally, binding *only* to the complex shifts the enzyme-*substrate* binding to the right (Le Chatelier's principle), leading to an increase in affinity and a **lowering of the apparent Michaelis constant K_M** by the same factor as the decrease in V_{max}.

The effects of inhibitors on enzyme kinetics can be visualized graphically. The Michaelis-Menten curve plots initial reaction velocity v_0 on the *y*-axis against the initial substrate concentration $[S]_0$ on the *x*-axis. With noncooperative enzymes, the result is a hyperbolic curve that approaches a maximal reaction velocity V_{max} when the enzyme is saturated. The K_M value can be visually approximated by identifying the substrate concentration that corresponds to a V_0 of ½ V_{max}. Only Choices A and B show the expected hyperbolic curve; of those two, only Choice B shows the smaller V_{max} and K_M expected for an uncompetitive inhibitor.

(Choice A) The inhibited curve approaches the same maximum value as the uninhibited curve (same V_{max}), but it takes more substrate to get there (higher K_M). Therefore, this plot represents competitive inhibition, not uncompetitive inhibition.

(Choices C and D) These graphs show linear plots that *may* be expected of a linearizing transformation like the Lineweaver-Burk plots. However, Lineweaver-Burk plots use the *reciprocal* parameters $1/V_0$ and $1/[S]_0$ as the *y*- and *x*-axes, respectively, instead of V_0 and $[S]_0$ as in a Michaelis-Menten graph. If the axes labels were altered accordingly, Choice C would represent purely noncompetitive inhibition, and Choice D would represent uncompetitive inhibition.

Educational objective:
Uncompetitive inhibition is a class of reversible inhibition that is characterized by the inhibitor's binding only the enzyme-substrate complex. With uncompetitive inhibition, enzymes show a decrease in both the apparent V_{max} and the apparent K_M.

Time spent:0 sec
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Enzymes